

Application form (Guide lines) for approval of M.Sc. in CSE Thesis proposal by the CASR. All the items, which are applicable of the following list, must be mentioned and filled in properly.

1. Name of the student: MD. SAKIF KHAN		Status: Part Time
Roll No: 0421052099		Session: APRIL-2021
2. Present Address: 404, Naddapara, Dakshinkhan, Dhaka		Cell: 01689960489
3. Name of Dept/Inst.: CSE		Programme: M.Sc. in CSE
4. Name of the Supervisor: Dr. Sadia Sharmin		Designation: Associate Professor
Supervisor Cell: +8801817108555		
5. Name of the Co-Supervisor:		Designation:
6. Date of First Enrollment in the Program: 01/04/2021		
7. Tentative Title:	AGENTIC GRAPH REASONING: AUTONOMOUS KNOWLEDGE GRAPH NAVIGATION FOR FACT VERIFICATION AND HALLUCINATION MITIGATION IN LARGE LANGUAGE MODELS	

Recent studies show that Large Language Models frequently hallucinate during complex tasks that require multi-step reasoning [1]. To fix this, researchers introduced Retrieval-Augmented Generation (RAG) [2], which connects the LLM to an external data sources. This allows the model to verify facts before answering.

However, standard RAG relies on vector similarity, which treats data as flat chunks of text. It often misses the complex relationships between entities [3]. Microsoft developed GraphRAG [4] using Knowledge Graphs (KGs) to provide global entity relationships. Although this method was useful, GraphRAG remains a static retrieval method.

Current publications now focus on Agentic AI, where the models do not just retrieve data but act upon it. At ACL (2025), the "Reasoning with Trees" framework showed that using search algorithms like Monte Carlo Tree Search allows an LLM to explore various reasoning paths within a graph [5]. HippoRAG (NeurIPS, 2024) introduced a neurobiologically inspired memory system [6]. This system uses a graph to help LLMs link facts over long conversations.

Researchers now use FactBench (2025) to measure accuracy [7]. Recent work on Multi-agent Debate [8] suggests that having multiple LLM agents cross-examine graph triples can identify inconsistencies. Furthermore, the Chain-of-Verification (CoVe) method [9] allows a model to fact-check its own reasoning paths against the ground truth of the graph.

For web data, which changes constantly, T-GraphRAG (2025) has introduced temporal awareness [10]. This allows agents to understand when a fact was last updated in a graph. Additionally, using Ontology-Guided Reasoning [11] ensures that the agent follows valid logical rules. This prevents the model from hallucinating non-existent relationships between entities.

Despite these efforts, models still struggle with multi-step reasoning. Present research at EDBT (2025) highlights a gap in autonomous self-correction [12]. We need agents that can backtrack and provide clear audit trails for web data management.

Our research goal is to develop an autonomous Agentic Graph Reasoning (AGR) framework that enhances trust in web data management by mitigating hallucinations.

There are mainly three components that cause LLMs to hallucinate: training data, model, and prompt. Our research addresses the knowledge gap at the prompt level. We'll provide the model with a structured context (via the KG), that can override the inherent errors in the model's training data.

We intend to achieve this goal by transforming static RAG into a dynamic navigation task. This study seeks to provide verifiable and explainable reasoning paths for complex, multi-hop queries. Our main goals for this study are as follows:

- i. To design an LLM-based agent using LangGraph capable of translating natural language into iterative graph queries (Cypher) to traverse complex entity relationships autonomously.
- ii. To develop a backtracking logic that identifies logical dead-ends within the graph and triggers recursive pathfinding to verify claims against ground-truth ontologies.
- iii. To build a verification layer that cross-references draft responses against ground-truth graph triples, forcing the model to autonomously correct its assertions when a logical inconsistency is detected.
- iv. To evaluate the framework's efficacy using FactBench and WebQSP datasets, measuring the reduction in hallucinations compared to standard GraphRAG baselines.

(b). Possible outcome:

The anticipated outcomes of this thesis are both technical and methodological. At the end of this study, we expect measurable increase in accuracy for queries requiring multi-step reasoning. Our proposed system will provide the exact traversal path used to generate each response. This would transform the LLM from a black box into a transparent, debugable reasoning engine. In addition we get a fact-verifying system which high-stakes industries, such as healthcare and finance data could adopt where ground truth are non-negotiables. We also expect to deliver an open-source, modular AGR implementation ready for enterprise or scientific data applications.

10. Outline of Methodology/ Experimental Design:

The research is organized into four phases:

Phase 1: Knowledge Environment Construction

The initial activity involves building a Knowledge Environment. I will connect the WebQSP and FactBench datasets into a Neo4j graph database. An LLM-based entity-relationship extractor will be used to ensure the graph's ontology (semantic alignment) is consistent with the target web data domains. This creates the ground truth skeleton required for the agent's navigation.

Phase 2: Agentic Architecture and Planning

I will develop the Agentic Reasoning Core using the LangGraph framework. This choice is critical as it supports the cyclic loops necessary for autonomous exploration. The agent will be designed with a ReAct (Reason + Act) loop, enabling it to generate dynamic Cypher queries to explore the graph in real-time. A central component of this phase is the implementation of a Path-Planning Module, which allows the agent to decompose a user's complex query into a series of atomic, searchable sub-questions.

Phase 3: The Navigation and Verification Loop

The core experimental logic involves a search-based navigation strategy inspired by Monte Carlo Tree Search (MCTS). As the agent traverses the graph, it will evaluate the relevance score of adjacent nodes. If a reasoning path leads to a factual dead-end, the agent is programmed to backtrack to the previous node and explore an alternative relationship. Before the final output is generated, a Structural Verification Layer will cross-reference the model's internal draft against the retrieved graph triples, forcing a self-correction loop if logical inconsistencies are found.

Phase 4: Evaluation and Benchmarking

The final activity is a comparative analysis against two baselines: standard Vector-RAG and static GraphRAG. The system will be evaluated on:

1. Factual Accuracy: Performance on multi-hop queries.
2. Hallucination Rate: Measured via the FactBench evaluator.
3. Path Fidelity: The degree of alignment between the agent's thought process and the actual graph structure.

11. a) Will you do the computational work through IICT computational facility and resources by online log in: No
b) If yes, then mention the total computational day: 0

12. References:

- [1] Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., and others, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," in *ACM Transactions on Information Systems*, Vol. 43, 2025, pp. 1-55 (2025).
- [2] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., and others, "Retrieval-augmented generation for knowledge-intensive nlp tasks," in **Advances in neural information processing systems**, Vol. 33, 2020, pp. 9459-9474 (2020).
- [3] Li, Y., Li, Z., Wang, P., Li, J., Sun, X., Cheng, H., and Yu, J. X., "A survey of graph meets large language model: Progress and future directions," in **arXiv preprint arXiv:2311.12399**, 2023, (2023).
- [4] Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., and Larson, J., "From local to global: A graph rag approach to query-focused summarization," in *arXiv preprint arXiv:2404.16130*, 2024, (2024).
- [5] Shen, T., Wang, J., Zhang, X., and Cambria, E., "Reasoning with trees: faithful question answering over knowledge graph," in *Proceedings of the 31st International Conference on Computational Linguistics*, 2025, pp. 3138-3157 (2025).
- [6] Gutiérrez, B. J., Shu, Y., Gu, Y., Yasunaga, M., and Su, Y., "Hipporag: Neurobiologically inspired long-term memory for large language models," in *Advances in neural information processing systems*, Vol. 37, 2024, pp. 59532-59569 (2024).
- [7] Bayat, F. F., Zhang, L., Munir, S., and Wang, L., "Factbench: A dynamic benchmark for in-the-wild language model factuality evaluation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 33090-33110 (2025).
- [8] Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., and Mordatch, I., "Improving factuality and reasoning in language models through multiagent debate," in *Forty-first international conference on machine learning*, 2024, (2024).
- [9] Dhuliawala, S., Komeili, M., Xu, J., Raileanu, R., Li, X., Celikyilmaz, A., and Weston, J., "Chain-of-verification reduces hallucination in large language models," in *Findings of the association for computational linguistics: ACL 2024*, 2024, pp. 3563-3578 (2024).
- [10] Li, D., Niu, Y., Ai, Y., Zou, X., Qi, B., and Liu, J., "T-grag: A dynamic graphrag framework for resolving temporal conflicts and redundancy in knowledge retrieval," in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 11880-11889 (2025).
- [11] Hu, Y., Xuan, W., Zhou, Q., Li, Z., Li, Y., Hu, J., and Fang, F., "A self-correcting Agentic Graph RAG for clinical decision support in hepatology," in *Frontiers in medicine*, Vol. 12, 2025, pp. 1716327 (2025).
- [12] Ma, C., Chen, Y., Wu, T., Khan, A., and Wang, H., "Unifying Large Language Models and Knowledge Graphs for Question Answering: Recent Advances and Opportunities," in *EDBT*, 2025, pp. 1174-1177 (2025).

13. List of courses so far taken with course no, name of the courses, credit hours, Grade, Grade Points and G. P. A (To be verified and signed by the Tabulator)

Course No	Course Title	Credit Hours	Grade	Grade Point	GPA
CSE 6506	Data Mining	3	B+	3	3.00
CSE 6302	Software Quality Assurance	3	A	3.5	
CSE 6305	Programming Languages and Systems	3	A+	4	
CSE 6708	Semantic Web	3	C	2	
CSE 6408	Advanced Algorithms	3	B+	3	
CSE 6406	Bioinformatics Algorithms	3	B	2.5	
CSE 6000	Thesis/Project	9	X	--	

Signature of the Tabulator: Abu Wasif

14. Cost Estimate: (Invoice / Quotation of material/equipment must be provided for every item which costs Tk. 10,000/- or more)		
a) Cost of Material (Breakup needed)	Tk.	0
b) Field works (if applicable)	Tk.	0
c) Conveyance/ Data Collection (With Breakup)	Tk.	0
d) Typing, Drafting, Binding & Paper etc.	Tk.	4500
e) Others (if any)	Tk.	0
Total	Tk.	4500

15. Approximate time (in hour) for BUET workshop facilities (if required): 0

16. Justification of having Co-Supervisor: N/A

17. Thesis/Project Proposal Approved by Doctoral Committee/BPGS/RAC reference:

Meeting no. BPGS/2026/04/21 Resolution No. 2b Date : 21/04/2026

18. Expected Pre-defense date: 02/05/2026 Pre-defense attended: 3

19. Expected Defense date: 06/06/2026

20. Time extension (if any):

- a) **1st Time extension (if any) up to:-----**
Meeting no.----- Resolution No.----- Date :-----
- b) **2nd Time extension (if any) up to:-----**
Meeting no.----- Resolution No.----- Date :-----
- c) **Further time extension (if any) up to:-----**
Meeting no.----- Resolution No.----- Date :-----

21. Appointment of Supervisor & Co-Supervisor Approved by the CASR Meeting No. (For Ph. D):

Meeting No.: Resolution No.: Date :

22. Appointment of Doctoral Committee Approved by the CASR Meeting No. (For Ph. D):

Meeting No.: Resolution No.: Date :

23. Result of the comprehensive examination for Ph. D (Photocopy of the result should be enclosed)

N/A

24. Number of Post-Graduate Student(s) working with the Supervisor at Present: 6

Number of PhD Student(s) working with the Supervisor at Present: 1

Md. Sakif Khan 06/07/2026 ----- Signature of the Student	Names and signatures of the members of the Doctoral Committee (if applicable)	
	1.	
Dr. Sadia Sharmin 06/07/2026 ----- Signature of the Supervisor	2.	
	3.	
	4.	
	5.	
Signature of Co-Supervisor	6.	
	7.	

Dr. Tanzima Hashem
30/04/2026

Signature of the Head/Director

Comments of the Head/Director
APPROVED.